

主 题:	ICASSP2025 : Paper 7271	
发件人:	"Microsoft CMT" <email@msr-cmt.org>	2024-11-12 10:34:35
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Shihong Duan has uploaded review for Martin: Mobility-Aware Reputation Mechanism for Federated Learning

-- Review Summary --

1. How confident are you in your evaluation of this paper?

2. Confident
2. Importance/Relevance

3. Of sufficient interest
3. Classify the type of paper

More empirical
4. Originality/Novelty

2. Minor originality; provides barely sufficient new insight or understanding
5. Justification of Originality/Novelty Score (required)

The problem posed in the paper is clear, and the solution demonstrates a certain level of innovation. However, the experimental data characteristics are simple, and the approach to addressing communication limitations is not sufficiently clear.
6. Theoretical Development

3. Probably correct; provides limited new insights or understanding
7. Experimental Validation

3. Limited but convincing
8. Clarity of Presentation

3. Clear enough
9. Reference to Prior Work

3. References adequate
10. Overall evaluation of this paper

2. Marginal reject
11. Justification of Overall evaluation of this paper (required)

1.The architecture of the paper is clear, but the color coding for the task steps is somewhat confusing. Currently, it appears that Step 4 is completed by the device, Step 1 involves the cloud, and the other task steps are handled by the edge server. The edge server bears a heavy computational load, so how scalable is this design? Additionally, the device selection criteria lack flexibility; how does the approach handle newly joined devices?
2.The experiment introduces three types of devices with different data quality (high-quality devices, data contamination devices, and sensor malfunction devices), which adequately considers the diversity and complexity of data quality in real IoT environments. Although the MNIST dataset is commonly used, its data characteristics are relatively simple. In the future, more complex datasets (such as CIFAR-10 or real IoT datasets) could be used to test the generalizability of the mechanism. Moreover, the result analysis could be more in-depth, including examining the impact of parameters on model performance.
3. The authors discuss communication limitations but lack analysis on how different bandwidth allocations affect device selection and model performance. It is necessary to include an analysis of model accuracy and convergence speed under different bandwidth conditions.
12. Award Quality (only for papers marked "Definite accept")

0: Not award quality
13. Confidential Comments to technical program committee (will not be seen by authors):

The paper addresses a real-world problem, but the test data used is relatively simple. Although the authors discuss communication limitations, there is a lack of analysis regarding model accuracy and convergence under different bandwidth conditions, which diminishes the value of the paper.

The proposed method places a substantial computational load on the edge server, and an analysis of the scalability of this design should be provided.

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